

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EC)

November 2013

EC304 Power Electronics

Date: 25.11.2013, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1(a) Enlist methods to turn off the thyristors. Explain any one in detail. [04]

(b) Choose appropriate answer: [03]

(I) Holding current of thyristor is :

- | | |
|----------------------------------|--------------------------------|
| (i) Less than latching current | (ii) Equal to latching current |
| (iii) More than latching current | (iv) Zero |

(II) SCR is a :

- | | |
|-------------------------------|---------------------------|
| (i) Unidirectional switch | (ii) Bidirectional switch |
| (iii) Four-directional switch | (iv) None of above |

(III) For Normal SCRs Switching turn on depends on :

- | | |
|---------------------|----------------------|
| (i) Gate Current | (ii) Cathode current |
| (iii) Anode Current | (iv) None of above |

Q - 2 (a) Enumerate types of power diodes. Explain schottky diode in brief. [04]

(b) Define reverse recovery time of power diodes and Compare Soft and Abrupt Recovery. [03]

(c) Explain Operating Modes of DC motor. [07]

OR

Q - 2 (a) Enlist advantages and disadvantages of IGBT. [04]

(b) What is the function of freewheeling diodes in controlled rectifier? [03]

(c) Draw block diagram for closed loop speed control of motor with inner current loop. [07]

Q - 3 (a) State the principle operation of inverter? [02]

(b) Derive equation for maximum transient voltage $V_{DT(max)}$ and steady state voltage $V_{DS(max)}$ across SCR in series connection of thyristors. [05]

(c) What are the different methods for voltage control of 1-Phase inverter? Explain any two. [07]

OR

- Q - 3 (a)** What is the dead zone in resonant inverter? [02]
- (b)** Compare symmetric and Asymmetric type of thyristors. [05]
- (c)** Explain the operation of 3-phase bridge inverter operating in 120° mode. Draw appropriate waveforms. [07]

SECTION – II

- Q - 4(a)** SCR cannot be triggered below 30° in 3-phase half wave Controlled Rectifier. True or False? Justify your answer. [03]
- (b)** Briefly discuss four quadrant chopper. [04]
- Q - 5 (a)** Explain Three phase semi converter in brief with necessary diagram and equation also sketch the Input and Output wave-form for (a) $\alpha = 15^\circ$ (b) $\alpha = 45^\circ$ [08]
- (b)** The boost regulator has an input voltage $V_s = 6V$. The average output voltage $V_a = 15V$ and average load current $I_a = 0.5\text{amp}$. Switching frequency is 20KHz , If $L = 250\mu\text{H}$ and $C = 440\mu\text{F}$. Determine (a) Duty Cycle (b) Ripple current of inductor ΔI (c) Peak current of inductor I_2 (d) Ripple voltage of capacitor ΔV_c . [06]

OR

- Q - 5 (a)** A half wave rectifier has a forward break down voltage $V = 150V$ when a gate current of 1mA flows in the gate circuit. If a sinusoidal voltage $V_s = 400V$ peak is applied. Find out. (i) Firing angle (ii) Average output voltage (iii) Average current for load resistance of 200Ω (iv) Power Output Assume that the gate current is 1mA throughout and the forward breakdown voltage is more than $400V$ when $I_g = 1\text{mA}$. [08]
- (b)** Discuss buck boost regulator with neat circuit diagram, necessary waveforms and equations. [06]
- Q - 6 (a)** Define snubber circuit. Enlist different types of it and discuss each with necessary detail. [06]
- (b)** Define Principle of phase control in case of AC voltage controller. Discuss 1- ϕ unidirectional AC voltage controller with necessary figure, waveform and equations. [08]

OR

- Q - 6 (a)** Write short note on fusing with neat sketch of fuse current. [06]
- (b)** An AC voltage controller has a resistive load $R = 10\Omega$ and RMS input voltage $V_s = 120V$, 60Hz . The thyristor switch is on for $n = 25$ cycle and is off for $m = 75$ cycle. (a) Determine RMS output voltage V_o , (b) The average current of thyristor. (c) The RMS current of thyristor. [08]
